Lecture 14. Non-parametric Tests

What are non-parametric tests?

- A class of statistical tests that make much weaker assumptions than their parametric equivalents (also called **assumption-free**).
- The term non-parametric applies to any test which doesn't assume the data are drawn from a specific distribution (also called **distribution-free**)
- We have already seen some examples of non-parametric tests:
  - Spearman's Rank correlation test for an association between two variables does not rely on normality.
  - The $\chi 2$ goodness of fit and the $\chi 2$ test of independence make weak assumptions about frequency data.

We use non the parametric version of a test when…

- our data violate the assumption of **normality**: A parametric test (t-tests, Person's r, and ANOVAs) usually includes some type of normality assumption, and, if that assumption is false, the conclusions drawn from that test may be inaccurate. In contrast, the validity of a non-parametric test is not affected by whether or not the variable is normally distributed in the population.
- our data include a few extreme scores (**outliers**): Since many NP tests rank the raw scores and operate on those ranks, they offer a test of differences in central tendency that are not affected by one or a few very extreme scores (outliers). An extreme score in a set of data can make the parametric test less powerful, because it inflates the variance, and hence the error term, as well as biasing the mean by shifting it toward the outlier (the latter may increase or decrease the mean difference).
- we are interested primarily in **medians** rather than means
- our **sample** size is too small to determine a normal distribution: In small datasets, it might be possible to reject the idea that they are normally distributed from plotting them but it is usually not possible to say with confidence that they are normally distributed (too few points to reliably get that information).

How do non-parametric tests work?

- Most work on the principle of **ranking** the data
  - Ranking: transforming raw scores into numbers that represent their position in an ordered list of those scores, from lowest (rank 1) to highest (rank N)
- The analysis is carried out **on the ranks** rather than the actual data
- The outcome of the analysis reasonably insensitive to the underlying distribution of the data

Parametric tests and their non-parametric alternatives

	Parametric	**Non-parametric**
Comparing two unpaired sample means	Two-Sample T-test	Wilcoxon rank-sum test / Mann-Whitey *U*-test
Comparing two paired sample means	Paired-Sample T-test	Wilcoxon signed-rank test
Comparing three or more group means	One-way ANOVA	Kruskal-Wallis test
Testing association between two measures	Pearson's Correlation (r)	Spearman's Rank Correlation (rho)

Between-groups: Wilcoxon rank-sum test

(Comparing two independent conditions)

Example: Do motor-speech skills differ between typically developing (TD) children and children with perinatal lesion (PL)?

- IV: group (TD/PL)
- Task: tongue twisters
- DV: number of errors

TD	PL
9	15
3	14
9	15
10	8
6	7
11	22
4	41
5	19
9	13
1	18
14	17

For this kind of study design, we should be able to run an independent-samples t-test. However, in this case we cannot do so, as the Shapiro-Wilk test of normality is significant, indicating that the data distribution is different from normal:

```
        Shapiro-Wilk normality test

data:  ttwdata %>% pull(PL)
W = 0.80901, p-value = 0.01236
```

Assumptions

Two-samples t-test:	**Wilcoxon rank-sum test:**
• Dependent variable is interval or ratio level • Two samples are independent • Dependent variable is normally distributed • Additional assumption for Student's t-test: variances of two samples are equal	• Dependent variable is ordinal (with more than 2 levels), interval, or ratio level • Two samples are independent

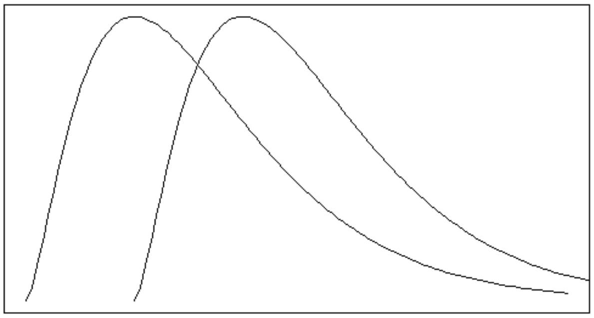
Hypotheses

- H_0: The two sets of scores were sampled from *identical* populations
- H_a: The two sets of scores were sampled from *different* populations

More specifically:

- H_0: the distributions of the two sets of scores *have the same central tendency (median)*
- H_a: the distributions of the two sets of scores *do not have the same central tendency (median)*

Identical distributions with different medians



Wilcoxon's rank-sum test evaluates the null hypothesis that the two sets of scores were sampled from identical populations. This is broader than the null hypothesis tested by the corresponding t-test, which dealt specifically with means (primarily as a result of the underlying assumptions that ruled out other sources of difference). If the two populations are assumed to have the same shape and dispersion, then the null hypothesis tested by the rank-sum test will actually deal with the central tendency (in this case the medians) of the two populations, and if the populations are also symmetric, the test will be a test of means. In any event, the rank-sum test is particularly sensitive to differences in central tendency.

Working out the ranks

We start by ranking the scores from lowest (2) to highest (41). The highest rank is N (the number of participants).

When multiple participants have the same score, we have to take the average of their ranks. In this example, we have two participants with a score of 15. Their ranks should be 16 and 17. We find the average: $\frac{16+17}{2} = 16.5$, so both participants get a rank of 16.5.

Three other participants scored 9 on the outcome measure – they should have ranks of 8, 9, and 10. The average of these ranks is $\frac{8+9+10}{3} = 9$, so all three participants get a rank of 9.

Raw Data

TD	PL
9	$\boxed{15}$
3	14
9	$\boxed{15}$
10	8
6	7
11	22
4	41
5	19
9	13
1	18
14	17

Ranked Data

TD	PL
9	$\boxed{16.5}$
2	14.5
9	$\boxed{16.5}$
11	7
5	6
12	21
3	22
4	20
9	13
1	19
14.5	18

Estimating the sum of ranks $\sum(R)$:

79.5 173.5

Estimating the *W* statistic

- Larger groups will have larger ranks (if unequal group sizes)
- Correcting by using the following formula:
  - mean rank = $\frac{N(N+1)}{2} = \frac{11(11+1)}{2} = 66$
- Calculating the potential values for W per group:
  - W = sum of ranks – mean rank
  - W_{TD}= 79.5 – 66 = **13.5**
  - W_{PL}= 173.5 – 66 = 107.5

The **smallest** of the two W values is our test statistic

- For sample sizes ≤ 10, check critical value from a table; for larger sizes use z-test
- No need to worry about that when using R!

The **interpretation** of W is qualitatively the same as the interpretation of a t-statistic. That is, if we want a two-sided test, then we reject the null hypothesis when W is very large or very small; but if we have a directional (i.e., one-sided) hypothesis, then we only use one or the other.

R uses the Monte Carlo method to obtain the significance level – this involves creating lots of data sets that match the sample but putting people from both groups randomly to either group. If the null hypothesis of no difference was true, the results would look very much like the data simulated with the Monte Carlo method, where group doesn't make a difference. An exact p-value is computed if the samples contain less than 50 finite values and there are no ties. Otherwise, a normal approximation is used. The normal approximation assumes that the sampling distribution of the W statistic is normal, which means that a standard error can be computed that is used to calculate a z score and an associated p-value.

Performing a Wilcoxon rank-sum test in R

TD	PL
9	15
3	14
9	15
10	8
6	7
11	22
4	41
5	19
9	13
1	18
14	17

````
```{r}
wxtest <- wilcox.test( ttwdata %>% pull(TD) , ttwdata %>% pull(PL) )
wxtest
```
````

```
	Wilcoxon rank sum test with continuity correction

data:  ttwdata %>% pull(TD) and ttwdata %>% pull(PL)
W = 13.5, p-value = 0.002224
alternative hypothesis: true location shift is not equal to 0
```

Continuity correction: Yates (1934) discussed in the χ^2 lecture last week.

Reporting the Wilcoxon rank-sum test

The data for the perinatal lesion group were not normally distributed (Shapiro-Wilk $p < .05$) so a Wilcoxon Rank Sum test was conducted to evaluate group differences in the tongue-twister task. Error rates in children with perinatal lesions (*Mdn* = 15) were significantly higher than those in typically developing children (*Mdn* = 9), W=13.5, n_1=11, n_2=11, p=.002.

Often this test is reported as a Mann-Whitney *U*-test:

... Error rates in children with perinatal lesions (*Mdn* = 15) were significantly higher than those in typically developing children (*Mdn* = 9), $\boldsymbol{U}$= 13.5, n_1=11, n_2=11, p=0.002.

Within-groups: Wilcoxon signed-rank test

Example: Can motor-speech skills improve with oral-motor exercises in children with perinatal lesion (PL)?

- IV: intervention (pre/post)
- Task: tongue twisters
- DV: number of errors

PLpre	PLpost
15	26
14	14
15	12
8	11
7	8
22	17
41	3
19	16
13	9
18	10
17	11

```
```{r}
ttwdata <- ttwdata %>% mutate(improvement = PLpre - PLpost)
shapiro.test(ttwdata %>% pull(improvement))
```

        Shapiro-Wilk normality test

data: ttwdata %>% pull(improvement)
W = 0.75213, p-value = 0.002216
```

Assumptions

Paired-samples t-test:	Wilcoxon signed-rank test:
• Dependent variable is interval or ratio level • Two samples are paired • Dependent variable (differences between conditions) is normally distributed	• Dependent variable is ordinal (with more than 2 levels), interval, or ratio level • Two samples are **paired**

Hypotheses

- H_0: the distribution of difference scores (in the population) is symmetric about zero.
- H_a: the distribution of difference scores (in the population) is not symmetric about zero.

More specifically:
- H_0: the median difference is zero / change in scores is zero / there is no change in scores after the intervention
- H_a: the median difference is not zero / change in scores is not zero / there is a change in scores after the intervention

This is the same hypothesis tested by the corresponding t-test when that test's normality assumption is met.

Ranking the data

First one needs to estimate the differences between the two conditions. We take note of the sign of the difference (positive or negative) and we rank the differences from smallest to largest ignoring the sign. If the difference is zero, we exclude these data from the ranking. We deal with tied scores the same way as we've done before. We then sum the positive differences (positive ranks) and negative ranks. Our test statistic T is the smaller of the two values.

PLpre	PLpost	Difference	Sign	Rank	Positive Ranks	Negative Ranks
15	26	-11	-	9		9
14	14	0	Exclude			
15	12	3	+	3	3	
8	11	-3	-	3		3
7	8	-1	-	1		1
22	17	5	+	6	6	
41	3	38	+	10	10	
19	16	3	+	3	3	
13	9	4	+	5	5	
18	10	8	+	8	8	
17	11	6	+	7	7	
				Total =	42	13

Performing a Wilcoxon signed-rank test in R

```
```{r}
wxsignedtest <- wilcox.test(ttwdata %>% pull(PL), ttwdata %>% pull(PLpost),
paired = TRUE, correct = FALSE )
wxsignedtest
```
```

```
	Wilcoxon signed rank test

data:  ttwdata %>% pull(PL) and ttwdata %>% pull(PLpost)
V = 42, p-value = 0.1384
alternative hypothesis: true location shift is not equal to 0
```

Reporting the Wilcoxon signed-rank test

For the Wilcoxon test we only need to report the significance of the test (and preferably the medians per condition):

> The perinatal lesion group's change scores from pre to post intervention were not normally distributed (Shapiro-Wilk $p < .05$) so a Wilcoxon Signed-Rank test was conducted to evaluate group change in tongue-twister task performance. Error rates before the intervention (*Mdn* = 15) were higher than after the intervention (*Mdn* = 11), but this difference was not significant (p=0.138).

Alternative to ANOVA: Kruskal-Wallis test

Example: Does lesion size (small, medium, large) affect motor-speech skills in children with perinatal lesion (PL)?

- IV: group (small, medium, large PL)
- Task: tongue twisters
- DV: number of errors

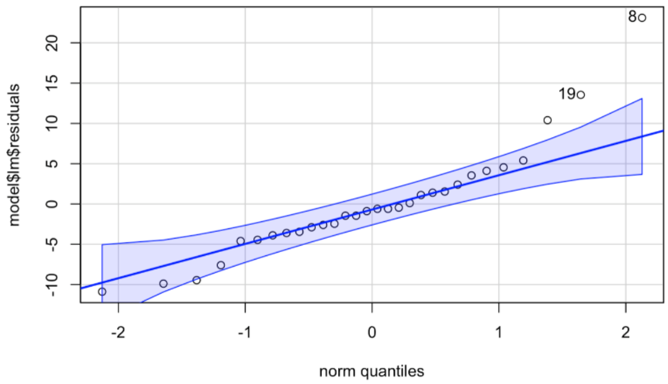
small	medium	large
11	11	8
4	14	14
19	17	22
1	10	18
8	9	17
10	12	15
14	11	7
8	26	41
6	8	19
5	3	16

Why not a one-way ANOVA?

```
```{r}
library(afex)
model <- aov_ez(id = "id", # the column containing the subject IDs
                dv = "scores", # the DV
                between = "LesionSize", # the between-subject variable
                data = lesdata)
anova(model)
```
```

```
        Shapiro-Wilk normality test

data:  model$lm$residuals
W = 0.89684, p-value = 0.007034
```



Assumptions

One-way ANOVA:	**Kruskal-Wallis:**
• The DV is interval or ratio level • Independent observations • Homogeneity variance of residuals • Normal distribution of residuals	• The DV is ordinal (with more than 2 levels), interval, or ratio level • Independent observations • Sample size: each group must have a sample size of 5 or more.

Hypotheses

- H_0: all samples are drawn from identical populations/ population distributions do not differ across groups.
- H_a: at least one sample is not drawn from identical populations/ population distributions do differ across groups.

More specifically:

- H_0: Scores will not differ for different lesion size groups
- H_a: Scores will differ for different lesion size groups

The Kruskal-Wallis test does not test specifically if the mean or median of the groups differ, only if the overall distributions are significantly different or not.

Analysis

Step 2: Prepare and rank your data – Arrange data from all groups together in one list in an ascending order – Give a rank to each of the data entries

Step 3: Sum the ranks for each group/condition

LesionSize	Score	Rank
small	1	1
medium	3	2
small	4	3
small	5	4
small	6	5
large	7	6
small	8	8.5
small	8	8.5
medium	8	8.5
large	8	8.5
medium	9	11
small	10	12.5
medium	10	12.5
small	11	15
medium	11	15
medium	11	15
medium	12	17
small	14	19
medium	14	19
large	14	19
large	15	21
medium	16	22
medium	17	23.5
large	17	23.5
large	18	25
small	19	26.5
large	19	26.5
large	22	28
medium	26	29
large	41	30

LesionSize	Score	Rank	Rank Sums
small	1	1	
small	4	3	
small	5	4	
small	6	5	
small	8	8.5	
small	8	8.5	
small	10	12.5	
small	11	15	
small	14	19	TotalSmall
small	19	26.5	103
medium	3	2	
medium	8	8.5	
medium	9	11	
medium	10	12.5	
medium	11	15	
medium	11	15	
medium	12	17	
medium	14	19	
medium	16	22	
medium	17	23.5	TotalMedium
medium	26	29	174.5
large	7	6	
large	8	8.5	
large	14	19	
large	15	21	
large	17	23.5	
large	18	25	
large	19	26.5	
large	22	28	TotalLarge
large	41	30	187.5

Calculating the test statistic H

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$$

where

k = the number of groups
n_i = the number of observations in group$_i$
R_i = the sum of the ranks in group$_i$
$N = \sum n_i$ = total sample size

To perform the Kruskal–Wallis test, we simply rank all scores without regard to group membership and then compute the sum of the ranks for each group. If the null hypothesis is true, we would expect the sums to be more or less equal (aside from difference due to the size of the samples). **H** = a measure of the degree to which the sums differ from one another.

Performing Kruskal-Wallis in R

````
```{r}
kruskal.test(Score ~ LesionSize, data=lesdata)
```
````

```
	Kruskal-Wallis rank sum test

data:  Score by LesionSize
Kruskal-Wallis chi-squared = 6.843, df = 2, p-value = 0.03266
```

Post-hoc pairwise comparisons

```
	Wilcoxon rank sum test with continuity correction

data:  phdata %>% pull(small) and phdata %>% pull(medium)
W = 32, p-value = 0.1119
alternative hypothesis: true location shift is not equal to 0


	Wilcoxon rank sum test with continuity correction

data:  phdata %>% pull(small) and phdata %>% pull(large)
W = 16, p-value = 0.01964
alternative hypothesis: true location shift is not equal to 0


	Wilcoxon rank sum test with continuity correction

data:  phdata %>% pull(medium) and phdata %>% pull(large)
W = 30.5, p-value = 0.1592
alternative hypothesis: true location shift is not equal to 0
```

Bonferroni correction:

p-value × number of comparisons

0.01964*3 = 0.05892 > alpha level

Reporting Kruskal-Wallis in R

The assumption of normality of the residuals was violated (Shapiro-Wilk $p < .05$) so a Kruskal-Wallis test was conducted to evaluate differences in the tongue-twister task across different lesion groups (small, medium, large). Error rates were significantly different across children with different perinatal lesion size, $H(2) = 6.84$, $p = .033$.

Children with large lesions ($Mdn = 17$) made more errors than children with medium ($Mdn = 11$) and small lesions ($Mdn = 8$). However, these differences were not significant when followed up with pairwise Wilcoxon's Rank Sum tests corrected for the number of comparisons.

Conclusion

Before we conclude, you might be wondering: Why not always run the non-parametric versions of tests and not bother with assumptions of the parametric ones?

Ranking data is a useful way around the distribution assumptions of parametric tests and outlier effects, however there is a price to pay: by ranking the data we lose some information about the **magnitude** of differences between scores. Consequently, non-parametric tests can be **less powerful** than their parametric counterparts. Reminder: power refers to the ability of a test to find an effect that genuinely exists. A parametric test is more likely to detect an effect than a non-parametric one. However, that is true only when the assumptions of the parametric test are met.